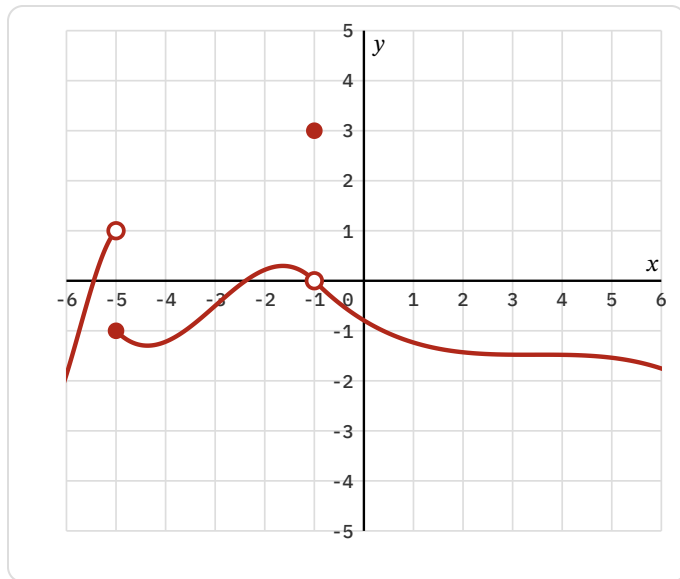


Mock Midterm B — Answer Key

Practice midterm (AI-Free) · Sections 1.1–1.4

Total: 95 points

1. Use the graph of $y = f(x)$ to state the value of each quantity, if it exists. If it does not exist, write DNE. 8 pts



- a.) $f(-5)$ (1)
-1
- b.) $\lim_{x \rightarrow -5^-} f(x)$ (1)
1
- c.) $\lim_{x \rightarrow -5^+} f(x)$ (1)
-1
- d.) $\lim_{x \rightarrow -5} f(x)$ (1)
DNE
- e.) $f(-1)$ (1)
3
- f.) $\lim_{x \rightarrow -1^-} f(x)$ (1)
0
- g.) $\lim_{x \rightarrow -1^+} f(x)$ (1)
0
- h.) $\lim_{x \rightarrow -1} f(x)$ (1)
0

WORKED SOLUTION

At $x = -5$

The **filled** dot at $x = -5$ sits at height -1, so $f(-5) = -1$.

Tracing the curve toward $x = -5$ from the left, the heights approach 1 (open or closed dot doesn't matter).

Tracing the curve toward $x = -5$ from the right, the heights approach -1 (open or closed dot doesn't matter).

The left and right limits don't match, so $\lim_{x \rightarrow -5} f(x)$ DNE.

At $x = -1$

The **filled** dot at $x = -1$ sits at height 3, so $f(-1) = 3$.

Tracing the curve toward $x = -1$ from the left, the heights approach 0 (open or closed dot doesn't matter).

Tracing the curve toward $x = -1$ from the right, the heights approach 0 (open or closed dot doesn't matter).

Both one-sided limits equal 0, so $\lim_{x \rightarrow -1} f(x) = 0$ — even though $f(-1) = 3$. The limit ignores the value AT the point.

2. Let $f(x) = \frac{\sqrt{x}-3}{x-9}$. We want to know what happens to the outputs when x is near **9**.

7 pts

x	8.9	8.99	8.999	9	9.001	9.01	9.1
$f(x)$	0.167132	0.166713	0.166671	?	0.166662	0.16662	0.166206

- a.) Can you find $\lim_{x \rightarrow 9} f(x)$ by plugging in $x = 9$? (1)

(B) No — substituting gives $\frac{0}{0}$, which is undefined

- b.) As $x \rightarrow 9^-$, $f(x)$ approaches... (2)

$$\frac{1}{6}$$

- c.) As $x \rightarrow 9^+$, $f(x)$ approaches... (2)

$$\frac{1}{6}$$

- d.) $\lim_{x \rightarrow 9} f(x) =$ (2)

$$\frac{1}{6}$$

WORKED SOLUTION

Plugging in $x = 9$ makes both the top and bottom 0, so we can't substitute.

Reading the table left to right, the outputs close in on $\frac{1}{6}$ from both sides (the values at 9 ± 0.001 are the closest).

$$\lim_{x \rightarrow 9} \frac{\sqrt{x}-3}{x-9} = \frac{1}{6}$$

Algebra check: $\frac{\sqrt{x}-3}{(\sqrt{x}-3)(\sqrt{x}+3)} = \frac{1}{\sqrt{x}+3} \rightarrow \frac{1}{6}$

3. Calculate each limit algebraically.

6 pts

a.) $\lim_{x \rightarrow -4} (2x^2 + 3x + 2)$ (2)
22

b.) $\lim_{x \rightarrow -3} \frac{x^2 - x - 9}{x - 2}$ (2)
 $-\frac{3}{5}$

c.) $\lim_{x \rightarrow 2} \sqrt{x^2 + 21}$ (2)
5

WORKED SOLUTION

Each input is in the domain of its function, so direct substitution works.

a. $\lim_{x \rightarrow -4} (2x^2 + 3x + 2) = 2(-4)^2 + 3(-4) + 2 = 22$

b. $\lim_{x \rightarrow -3} \frac{x^2 - x - 9}{x - 2} = \frac{(-3)^2 - 1(-3) - 9}{-3 - 2} = \frac{3}{-5} = -\frac{3}{5}$

c. $\lim_{x \rightarrow 2} \sqrt{x^2 + 21} = \sqrt{(2)^2 + 21} = \sqrt{25} = 5$

4. Calculate each limit algebraically. (Direct substitution gives $\frac{0}{0}$ — simplify first.) If it doesn't exist, write DNE. 4 pts

a.) $\lim_{x \rightarrow 16} \frac{x - 16}{\sqrt{x} - 4}$ (2)
8

b.) $\lim_{h \rightarrow 0} \frac{(2+h)^3 - 8}{h}$ (2)
12

WORKED SOLUTION

a.

$$\lim_{x \rightarrow 16} \frac{x - 16}{\sqrt{x} - 4} = \lim_{x \rightarrow 16} \frac{(\sqrt{x} - 4)(\sqrt{x} + 4)}{\sqrt{x} - 4} = \lim_{x \rightarrow 16} (\sqrt{x} + 4) = 8$$

b.

$$(2 + h)^3 = 8 + 12h + 6h^2 + h^3, \text{ so } \lim_{h \rightarrow 0} \frac{(2 + h)^3 - 8}{h} = \lim_{h \rightarrow 0} (12 + 6h + h^2) = 12$$

Notation tip: keep writing **lim** on every line until the step where you actually substitute.

5. Compute each limit algebraically. Use $\infty / -\infty$ for unbounded results.

6 pts

a.) $\lim_{x \rightarrow -\infty} \frac{4x^2+1}{6x^2-9}$ (2)
 $\frac{2}{3}$

b.) $\lim_{x \rightarrow \infty} \frac{4}{-x^4-7x^2}$ (2)
 0

c.) $\lim_{x \rightarrow \infty} \frac{x^5-7x^4+7x^3-3x^2+7x+4}{-4x^3+9x^2}$ (2)
 $-\infty$

WORKED SOLUTION

a. Divide every term by x^2 , the highest power of x in the denominator:

$$\lim_{x \rightarrow -\infty} \frac{4x^2+1}{6x^2-9} = \lim_{x \rightarrow -\infty} \frac{4 + \frac{1}{x^2}}{6 - \frac{9}{x^2}}$$

Every fraction term $\rightarrow 0$, leaving $\frac{4}{6} = \frac{2}{3}$.

b. Divide every term by x^4 , the highest power of x in the denominator:

$$\lim_{x \rightarrow \infty} \frac{4}{-x^4-7x^2} = \lim_{x \rightarrow \infty} \frac{\frac{4}{x^4}}{-1 - \frac{7}{x^2}}$$

The top $\rightarrow 0$ while the bottom $\rightarrow -1$, so the limit is 0 .

c. Divide every term by x^3 , the highest power of x in the denominator:

$$\lim_{x \rightarrow \infty} \frac{x^5-7x^4+7x^3-3x^2+7x+4}{-4x^3+9x^2} = \lim_{x \rightarrow \infty} \frac{x^2-7x+7-\frac{3}{x}+\frac{7}{x^2}+\frac{4}{x^3}}{-4+\frac{9}{x}}$$

The bottom $\rightarrow -4$, but the top is dominated by $1x^2$, which $\rightarrow \infty$ as $x \rightarrow \infty$; dividing by -4 gives $-\infty$.

6. Let

7 pts

$$G(x) = \begin{cases} -x^2 & x < -1 \\ 2x + 1 & x > -1 \end{cases}$$

Find each quantity. If it does not exist, write DNE.

a.) $G(-1)$ (1)

DNE

b.) $\lim_{x \rightarrow -1^-} G(x)$ (1)

-1

c.) $\lim_{x \rightarrow -1^+} G(x)$ (1)

-1

d.) $\lim_{x \rightarrow -1} G(x)$ (1)

-1

e.) $G(2)$ (1)

5

f.) $\lim_{x \rightarrow -\infty} G(x)$ (1)

$-\infty$

g.) $\lim_{x \rightarrow \infty} G(x)$ (1)

∞

WORKED SOLUTION

$G(-1)$: the pieces cover $x < -1$ and $x > -1$ only, so $G(-1)$ is undefined $\rightarrow$ DNE.

Left: use $-x^2$: $-(-1)^2 + 0 = -1$.

Right: use $2x + 1$: $2(-1) + 1 = -1$.

Two-sided: both sides give -1, so the limit is -1 (even though $G(-1)$ is undefined).

$G(2)$: $2 > -1$, so use $2x + 1$: $2(2) + 1 = 5$.

End behavior: as $x \rightarrow -\infty$ we're on the $-x^2$ piece — an even-degree term with negative leading coefficient $\rightarrow -\infty$. As $x \rightarrow \infty$ we're on the line with slope 2 $\rightarrow \infty$.

7. Let

6 pts

$$f(x) = \begin{cases} \frac{x^2 - 2x - 8}{x + 2} & x < -2 \\ 3x + 2 & x \geq -2 \end{cases}$$

Find each limit or function value algebraically. Write DNE if it doesn't exist.

a.) $\lim_{x \rightarrow 0} f(x)$ (1)
2

b.) $\lim_{x \rightarrow -4} f(x)$ (1)
-8

c.) $f(-2)$ (1)
-4

d.) $\lim_{x \rightarrow -2^-} f(x)$ (1)
-6

e.) $\lim_{x \rightarrow -2^+} f(x)$ (1)
-4

f.) $\lim_{x \rightarrow -2} f(x)$ (1)
DNE

WORKED SOLUTION

For $x < -2$, factor the top: $\frac{(x+2)(x-4)}{x+2} = x - 4$ (valid since $x \neq -2$ there).

a. $0 > -2 \rightarrow$ bottom piece: $3(0) + 2 = 2$.

b. $-4 < -2 \rightarrow$ top piece, simplified: $-4 - 4 = -8$.

c. The bottom piece includes $x = -2$ ($\geq$), so $f(-2) = 3(-2) + 2 = -4$.

d. Left: $\lim_{x \rightarrow -2^-} (x - 4) = -6$.

e. Right: $\lim_{x \rightarrow -2^+} (3x + 2) = -4$.

f. $-6 \neq -4$, so the limit DNE.

8. Find the value of k that makes f continuous at $x = -2$.

4 pts

$$f(x) = \begin{cases} x^2 + x & x < -2 \\ kx + 6 & x \geq -2 \end{cases}$$

a.) $\lim_{x \rightarrow -2^-} f(x)$

(2)

2

b.) $k =$

(2)

2

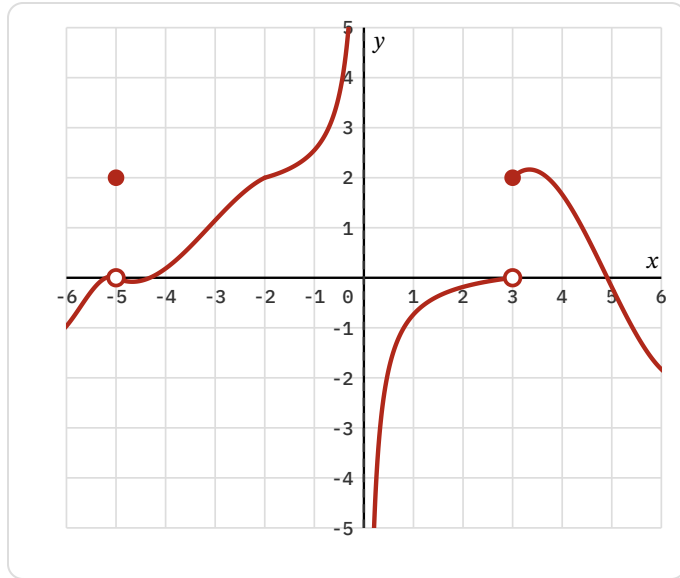
WORKED SOLUTION

Left: $(-2)^2 + 1(-2) = 2$. Right (and $f(-2)$): $k(-2) + 6$.

$$-2k + 6 = 2 \Rightarrow k = \frac{-4}{-2} = 2$$

9. Classify the behavior of f at each labeled x -value.

4 pts



- a.) At $x = -5$ (1)
(B) Removable discontinuity
- b.) At $x = -2$ (1)
(A) Continuous
- c.) At $x = 0$ (1)
(D) Infinite discontinuity
- d.) At $x = 3$ (1)
(C) Jump discontinuity

WORKED SOLUTION

$x = -5$: Both sides approach 0, but the filled dot is at 2 $\rightarrow$ removable (the limit exists, it just doesn't match f).

$x = -2$: No break, and the dot is on the curve $\rightarrow$ continuous.

$x = 0$: The graph shoots off along a vertical asymptote $\rightarrow$ infinite.

$x = 3$: Left approaches 0, right approaches 2 $\rightarrow$ jump.

10. Let $C(q) = 3q^2 + 2q + 163$, where $C(q)$ represents the cost, in dollars, of producing q items. 7 pts
Find the average rate of change over $[1, 5]$.

a.) $C(1)$ (2)

168

b.) $C(5)$ (2)

248

c.) Average rate of change (number only) (2)

20

d.) Which sentence correctly states the result **with units**? (1)

(A) Between $q = 1$ and $q = 5$ items, the cost changes at an average rate of 20 dollars per item.

WORKED SOLUTION

$$C(1) = 168, \quad C(5) = 248$$

$$\frac{C(5) - C(1)}{5 - 1} = \frac{248 - (168)}{4} = \frac{80}{4} = 20 \text{ dollars per item}$$

It's an *average* over an interval (not an instant), and the units are dollars per item — dollars per item. The total change would be 80 dollars.

11. Let $f(x) = \frac{6}{x}$. Calculate the average rate of change of f over each interval (rounded answers to 2+ decimals are fine). 5 pts

a.) $[2, 5]$ (1)
 $-\frac{3}{5}$

b.) $[2, 3]$ (1)
 -1

c.) $[2, 2.5]$ (1)
 $-\frac{6}{5}$

d.) $[2, 2.05]$ (1)
 $-\frac{60}{41}$

e.) As the interval shrinks toward $x = 2$, these values approach... (1)
 $(A) -\frac{3}{2}$

WORKED SOLUTION

$$[2, 5] : \frac{\frac{6}{5} - 3}{3} = -\frac{3}{5} \approx -0.6$$

$$[2, 3] : \frac{2 - 3}{1} = -1$$

$$[2, 2.5] : \frac{\frac{12}{5} - 3}{0.5} = -\frac{6}{5} \approx -1.2$$

$$[2, 2.05] : \frac{\frac{120}{41} - 3}{0.05} = -\frac{60}{41} \approx -1.4634$$

Each value is the slope of a secant line through $(2, 3)$. As the second point slides toward the first, the secant slopes approach the slope of the **tangent line**:

$$-\frac{6}{2^2} = -\frac{3}{2} \text{ — the instantaneous rate of change at } x = 2.$$

12. Let $f(x) = x^3 - 3x$. Find and simplify the difference quotient $\frac{f(x+h)-f(x)}{h}$, then use it to fill 7 pts in the table.

a.) Simplified difference quotient (in terms of x and h) (3)

$$3x^2 + 3xh + h^2 - 3$$

b.) $x = -1, h = 1$ (1)

$$-2$$

c.) $x = -1, h = 0.1$ (1)

$$-\frac{29}{100}$$

d.) $x = -1, h = 0.01$ (1)

$$-\frac{299}{10000}$$

e.) As $h \rightarrow 0$, what value do these approach? (1)

$$0$$

WORKED SOLUTION

$$f(x+h) = (x+h)^3 - 3(x+h) = x^3 + 3x^2h + 3xh^2 + h^3 - 3x - 3h$$

$$\frac{f(x+h) - f(x)}{h} = \frac{3x^2h + 3xh^2 + h^3 - 3h}{h} = 3x^2 + 3xh + h^2 - 3$$

Now substitute:

$$x = -1, h = 1 : -2$$

$$x = -1, h = 0.1 : -\frac{29}{100} = -0.29$$

$$x = -1, h = 0.01 : -\frac{299}{10000} = -0.0299$$

Letting $h \rightarrow 0$ leaves 0: the instantaneous rate of change (slope of the tangent line) at $x = -1$. That's exactly what section 1.4 formalizes.

13. Let $f(x) = 3x^2 - 2x - 6$. Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$. 8 pts

a.) $f'(x) =$ (3)

$$6x - 2$$

b.) $f'(-3) =$ (2)

$$-20$$

c.) Slope of the tangent line to f at $x = 2$ (2)

$$10$$

d.) What does $f'(-3) = -20$ mean? (1)

(D) The instantaneous rate of change of f at $x = -3$ is -20 — the slope of the tangent line there.

WORKED SOLUTION

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{[3(x+h)^2 - 2(x+h) - 6] - (3x^2 - 2x - 6)}{h} \\ &= \lim_{h \rightarrow 0} \frac{3x^2 + 6xh + 3h^2 - 2x - 2h - 6 - 3x^2 + 2x + 6}{h} \\ &= \lim_{h \rightarrow 0} \frac{6xh + 3h^2 - 2h}{h} = \lim_{h \rightarrow 0} \frac{h(6x + 3h - 2)}{h} \\ &= \lim_{h \rightarrow 0} (6x + 3h - 2) = 6x - 2 \end{aligned}$$

Then $f'(-3) = -20$ and the slope of the tangent at $x = 2$ is $f'(2) = 10$.

$f'(-3)$ is the *instantaneous* rate of change at one input — the slope of the tangent line, not the height of the graph and not an average over an interval.

14. Let $f(x) = x^2 - 8x - 3$, so $f'(x) = 2x - 8$.

6 pts

a.) At what x is the tangent line horizontal?

(2)

4

b.) What is the y -coordinate of that point?

(2)

-19

c.) At what x does the tangent line have slope 2?

(2)

5

WORKED SOLUTION

$$f'(x) = 0 : \quad 2x - 8 = 0 \Rightarrow x = 4$$

$$f(4) = -19$$

$$f'(x) = 2 : \quad 2x - 8 = 2 \Rightarrow x = \frac{10}{2} = 5$$

15. Let $h(t) = -16t^2 + 128t$ be the height, in feet, of a rocket t seconds after launch.

6 pts

a.) $h'(t) =$ (3)
 $-32t + 128$

b.) $h'(5)$ (2)
 -32

c.) Interpret $h'(5) = -32$. (1)
(C) At $t = 5$ seconds, the height of the rocket is decreasing at a rate of 32 feet per second.

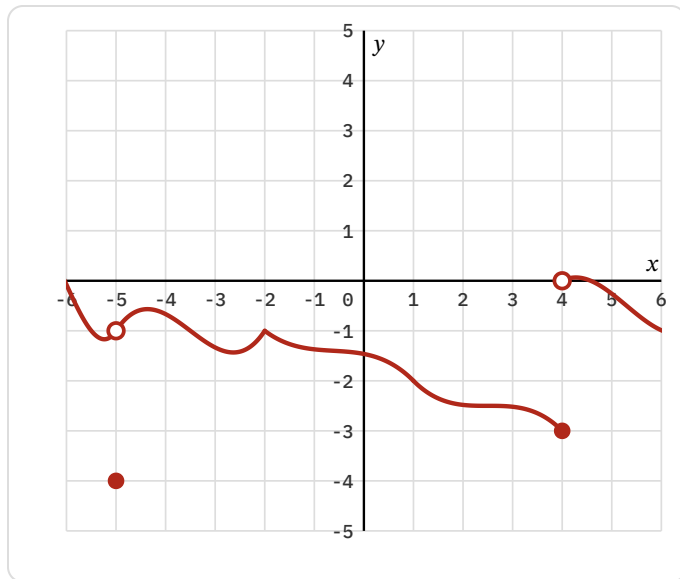
WORKED SOLUTION

$$\begin{aligned} h'(t) &= \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h} = \lim_{h \rightarrow 0} \frac{[-16(t+h)^2 + 128(t+h)] - (-16t^2 + 128t)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-16t^2 - 32th - 16h^2 + 128t + 128h + 16t^2 - 128t}{h} \\ &= \lim_{h \rightarrow 0} \frac{-32th - 16h^2 + 128h}{h} = \lim_{h \rightarrow 0} \frac{h(-32t - 16h + 128)}{h} \\ &= \lim_{h \rightarrow 0} (-32t - 16h + 128) = -32t + 128 \end{aligned}$$

$$h'(5) = -32(5) + 128 = -32$$

Positive/negative tells the direction, and the units are feet per second: at that instant the height of the rocket is decreasing at 32 feet per second. (The value $h(5) = 240$ is a different quantity.)

16. At each labeled x -value, decide whether the graph of f is continuous there and whether it is differentiable there. 4 pts



- a.) At $x = -5$ (1)
(C) Not continuous (so not differentiable)
- b.) At $x = -2$ (1)
(B) Continuous but NOT differentiable
- c.) At $x = 1$ (1)
(A) Continuous and differentiable
- d.) At $x = 4$ (1)
(C) Not continuous (so not differentiable)

WORKED SOLUTION

$x = -5$: There's a hole (the dot isn't on the curve) $\rightarrow$ not continuous, so not differentiable.

$x = -2$: No break, but the curve comes to a sharp point (corner): the slopes from the left and right disagree $\rightarrow$ continuous, not differentiable.

$x = 1$: Smooth and unbroken — a single tangent line fits $\rightarrow$ continuous and differentiable.

$x = 4$: The graph jumps $\rightarrow$ not continuous, so not differentiable.
